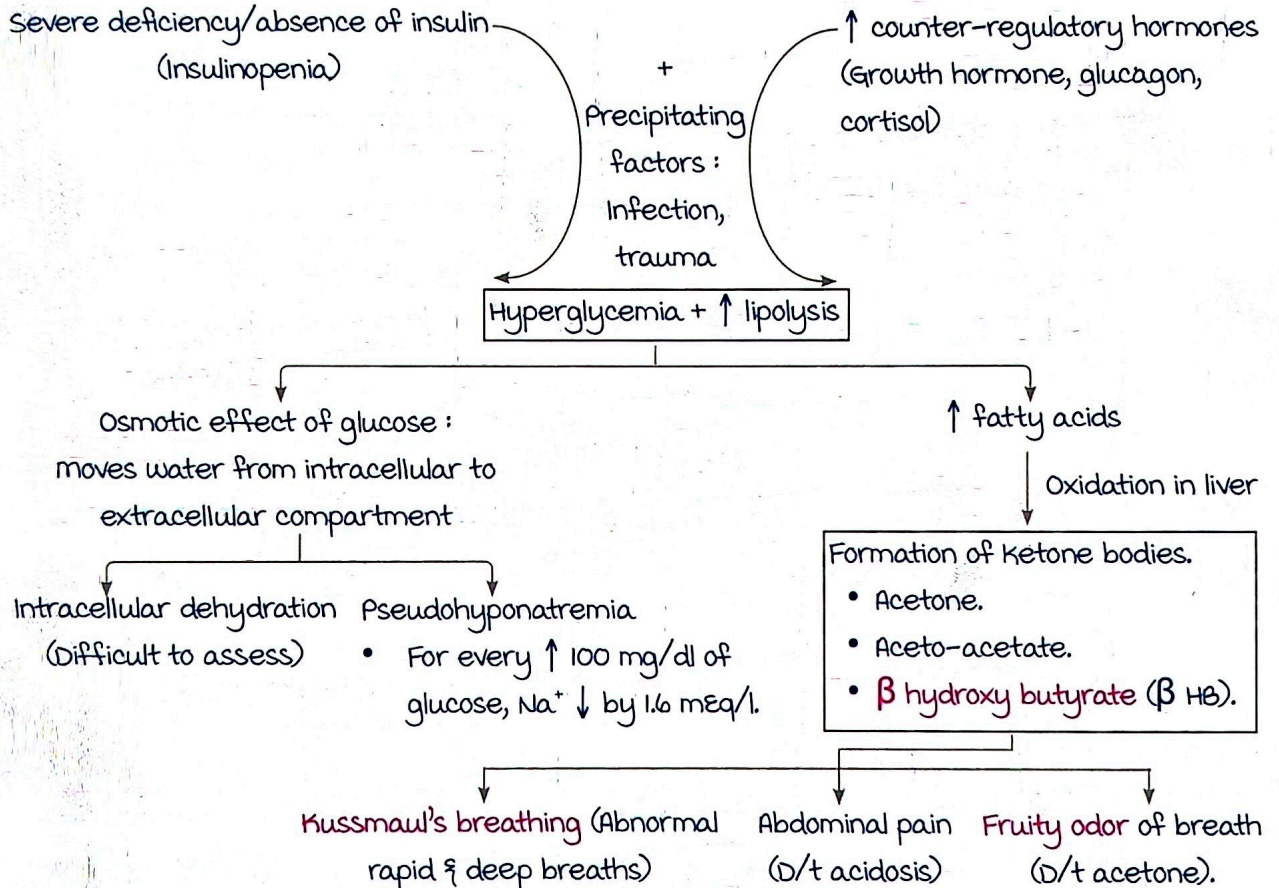


----- Active space -----

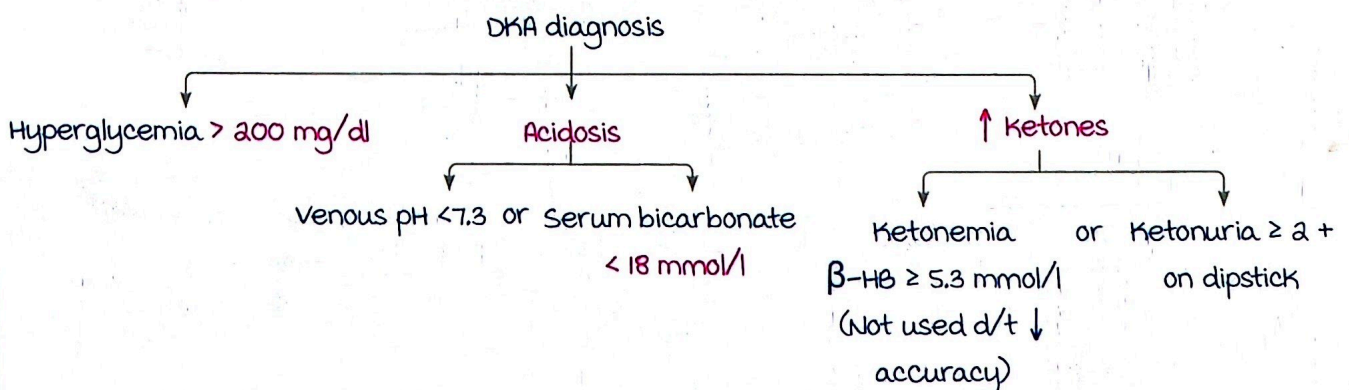
DIABETIC KETOACIDOSIS

Based on International Society for Pediatric and Adolescent Diabetes (ISPAD) guidelines.

PATHOPHYSIOLOGY:



CRITERIA:



Diabetic Ketoacidosis

213

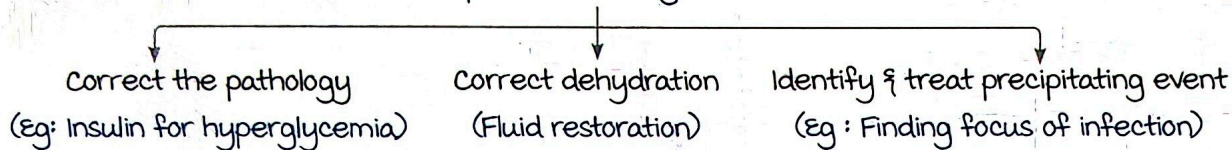
Severity of DKA :

----- Active space -----

	mild DKA	moderate DKA	Severe DKA
venous pH	7.2 - 7.3	7.1 - 7.2	< 7.1
Serum bicarbonate	< 18 mmol/L	< 10 mmol/L	< 5 mmol/L
Estimated dehydration	5 % of body weight (BW)	7 % of BW	10 % of BW

MANAGEMENT

3 pillars of management



Initial management :

	minimal dehydration	moderate or greater dehydration	Shock
Features	Tolerates oral fluids	- Not in shock. - Intolerant to oral fluids. - Symptoms of dehydration + vomiting.	- Thready pulses. ↓ consciousness or coma.
Fluid therapy (Based on estimated dehydration)	Oral fluids	10 - 20 ml/kg of IV NS (0.9 %) over 20-30 minutes	- Airway, breathing, circulation secured. - IV bolus of 20 ml/kg 0.9 % NS.
Insulin therapy	s/c insulin	- Start after the fluid therapy. - IV infusion (0.05 - 0.1 U/kg/hr) $\xrightarrow[\text{improvement}]{\text{upon}}$ s/c insulin.	

C/I : Insulin bolus (↑ ↑ insulin during resuscitation → Cerebral edema).

Further management

Presentation	management
Cerebral edema	Slow deficit correction over 24 - 48 hr
Insulin Induced hypokalemia	- NS + 40 meq/L of KCl used - C/I : Bicarbonate therapy (D/t worsening of CNS acidosis & hypokalemia).
Blood glucose < 300 mg/dl	5 % dextrose added to IV fluids (To prevent hypoglycemia)
Hypernatremia	Restores on treating hyperglycemia